

Enthusiast, Leader & Achiever Course

PHASE : (All Phase)

TARGET : PRE-MEDICAL 2020

Test Type : MAJOR

Test Pattern : NEET (UG)

TEST DATE : 03-05-2020

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	1	3	3	3	3	4	1	1	4	4	1	4	3	3	2	3	4	1	3	3	4	3	2	3	3	3	2	4	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	4	1	1	1	3	2	1	2	2	2	3	1	4	3	2	2	1	4	2	4	2	3	1	2	1	2	2	4	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	2	1	4	1	1	3	2	2	1	4	3	2	4	1	1	3	1	3	3	1	3	2	2	2	3	3	1	2	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	2	1	3	1	4	1	1	2	1	3	3	1	2	2	1	4	3	3	3	4	4	2	1	2	2	3	1	2	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	2	2	3	4	1	3	3	2	3	1	4	1	4	1	3	4	1	3	2	1	3	4	3	3	2	2	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	2	2	3	3	4	2	1	2	3	1	2	1	2	4	4	3	3	2	1	2	2	3	3	3	1	3	2	3	4

HINT - SHEET

2. Ans (1)

$$w = \frac{1}{2} \times 6 \times 10 - 4 \times 5 + 4 \times 5 - 5 \times 2$$

$$= 30 - 20 + 20 - 10 = 20\text{J}$$

$$w = k_f - k_i$$

$$20 = k_f - 25$$

$$k_f = 45\text{J}$$

3. Ans (3)

$$\frac{U_1}{U_2} = \frac{\frac{1}{2}k_1x_1^2}{\frac{1}{2}k_2x_2^2} = \frac{(k_1x_1)^2}{k_1} \times \frac{k_2}{(k_2x_2)^2}$$

$$= \frac{F^2}{k_1} \times \frac{k_2}{F^2} = \frac{k_2}{k_1} = \frac{3000}{1500} = 2 : 1$$

4. Ans (3)

$$E_i = E_f \Rightarrow \frac{1}{2}mv^2 = mg(R - h)$$

$$\Rightarrow v = \sqrt{2g(R - h)}$$

6. Ans (3)

$$\frac{dp}{dt}$$

7. Ans (4)

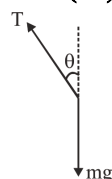
$$\text{Radius of gyration } K = \sqrt{\frac{I}{m}}$$

$$K_{\text{disc}} = \sqrt{\frac{\frac{1}{2}mR^2 + mR^2}{m}} = \sqrt{\frac{3}{2}} R$$

$$K_{\text{ring}} = \sqrt{\frac{mR^2 + mR^2}{m}} = \sqrt{2} R$$

$$\therefore \frac{K_{\text{disc}}}{K_{\text{ring}}} = \frac{\sqrt{3/2}}{\sqrt{2}} = \frac{\sqrt{3}}{2}$$

8. Ans (1)



$$T \cos \theta = mg$$

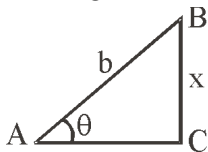
$$T \sin \theta = m\omega^2 r$$

$$\Rightarrow T \sin \theta = m l \sin \theta \omega^2$$

$$\Rightarrow T = m l \omega^2$$

9. **Ans (1)**

As shown in the figure, let θ be the angle of banking to counteract the centrifugal force.



$$\text{Now, } \tan \theta = \frac{v^2}{Rg}$$

If x be the elevation required then,

$$\sin \theta = \frac{x}{b} \text{ or } \theta = \sin \theta = \tan \theta = \frac{x}{b}$$

$$\therefore \frac{x}{b} = \tan \theta = \frac{v^2}{Rg}$$

$$\text{or } x = \frac{v^2 b}{Rg}$$

11. **Ans (4)**

$$\text{Given } s = t^3 + 5$$

$$V = 3t^2$$

$$a_t = 6t$$

$$\text{and } a_{cp} = \frac{(3t^2)^2}{R}$$

$$\text{at } t = 2s \quad a_t = 12 \text{ ms}^{-2}$$

$$a_{cp} = \frac{9(16)}{20} = \frac{36}{5} \text{ ms}^{-2}$$

$$\therefore a_{\text{net}} = \sqrt{a_t^2 + a_{cp}^2} = \sqrt{(12)^2 + \left(\frac{36}{5}\right)^2}$$

$$= 13.994 \approx 14 \text{ ms}^{-2}$$

12. **Ans (1)**

$$\therefore \vec{R}_{\text{cm}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

$$= \frac{10(\hat{i} + \hat{j} + \hat{k}) + 30(-\hat{i} - \hat{j} - \hat{k})}{10 + 30}$$

$$= \frac{10\hat{i} + 10\hat{j} + 10\hat{k} - 30\hat{i} - 30\hat{j} - 30\hat{k}}{40}$$

$$= \frac{-20\hat{i} - 20\hat{j} - 20\hat{k}}{40}$$

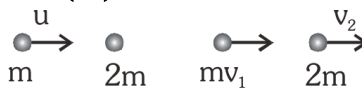
$$\vec{R}_{\text{cm}} = \frac{-(\hat{i} + \hat{j} + \hat{k})}{2}$$

13. **Ans (4)**

$$m_1 = 2\text{kg}, m_2 = 4\text{kg}, \vec{v}_1 = 20 \text{ m/s}, \vec{v}_2 = -10 \text{ m/s}$$

$$\vec{V}_{\text{cm}} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} = \frac{2 \times 20 - 4 \times 10}{2 + 4} = 0 \text{ m/s}$$

16. **Ans (2)**



$$\text{before} \quad u = v_1 + 2v_2 \quad \dots(1)$$

$$e = 1 = \frac{-(v_1 - v_2)}{u}$$

$$\therefore u = v_2 - v_1 \quad \dots(2)$$

$$\text{solving } v_1 = \frac{-u}{3}, v_2 = \frac{2u}{3}$$

fraction energy loss of neutron

$$\frac{\frac{1}{2}mu^2 - \frac{1}{2}m\left(-\frac{u}{3}\right)^2}{\frac{1}{2}mu^2} = \frac{8}{9}$$

17. **Ans (3)**

For constant acceleration

$$n_1 : n_2 : n_3 \dots = 1 : 3 : 5 : \dots$$

$$\therefore \text{if } n_1 = 10 \text{ rotation then } n_2 = 30 \text{ rotation}$$

19. **Ans (1)**

$$\therefore \tau = I\alpha \quad \therefore (mg) \frac{\ell}{2} = \left(\frac{m\ell^2}{3}\right) \alpha \Rightarrow \alpha = \frac{3g}{2\ell}$$

21. **Ans (3)**

$$I_1 \omega_1 = I_2 \omega_2 \text{ by COAM}$$

$$\omega_2 = \frac{I_1 \omega_1}{I_2}$$

$$I_1 = MR^2$$

$$\omega_1 = \omega$$

$$I_2 = MR^2 + mR^2 + mR^2 = (M + 2m)R^2$$

$$\omega_2 = \frac{M\omega}{M + 2m}$$

23. **Ans (3)**

$$v = \sqrt{2as} = \sqrt{2 \times \left(\frac{5}{7}\right) gl \sin \theta} = \sqrt{\frac{10}{7} gl \sin \theta}$$

$$\text{Hints : } a = \frac{g \sin \theta}{1 + \frac{k^2}{R^2}}$$

$$Mk^2 = \frac{2}{5}MR^2 \Rightarrow \frac{k^2}{R^2} = \frac{2}{5}$$

24. **Ans (2)**

Given : $P_1 = P$, $P_2 = 3P$

$$\frac{\Delta K}{K} = \left[\left(\frac{P_2}{P_1} \right)^2 - 1 \right] \times 100$$

$$= \left[\left(\frac{3P}{P} \right)^2 - 1 \right] \times 100$$

$$\frac{\Delta K}{K} = 800\%$$

26. **Ans (3)**

Weight of body on the surface of the earth = $mg = 72 \text{ N}$

Acceleration due to gravity at height h is

$$g_h = \frac{gR_E^2}{(R_E + h)^2} = \frac{gR_E^2}{\left(R_E + \frac{R_E}{2}\right)^2} = \frac{4}{9}g$$

$$\left(\because h = \frac{R_E}{2} \right)$$

Gravitational force on body at height h is

$$F = mg_h =$$

$$m \times \frac{4}{9}g = \frac{4}{9} \times mg = \frac{4}{9} \times 72 \text{ N} = 32 \text{ N}$$

28. **Ans (2)**

$$v_e = \sqrt{2Rg}$$

$$g = \frac{4}{3}\pi R\rho G$$

$$v_c = \sqrt{2R \times \frac{4}{3}\pi R\rho G}$$

$$v_e \propto R\sqrt{\rho}$$

30. **Ans (4)**

Fact

31. **Ans (2)**

$$F = yA \propto \Delta\theta = 10^{11} \times 10^{-4} \times 10^{-5} \times 100 = 10^4 \text{ N}$$

32. **Ans (4)**

When the sphere falls in a gravity-free region, no force acts on it in any direction initially. Hence, its terminal velocity will be zero, i.e., it will remain at rest.

33. **Ans (1)**

$$4(H - 4) = 6(H - 6)$$

$$\text{or } 2H = 36 - 16 = 20 \text{ or } H = 10 \text{ cm}$$

34. **Ans (1)**

$$T = \frac{W}{\Delta A} = \frac{2 \times 10^{-4}}{2 \times (50 \times 10^{-4})} = 2 \times 10^{-2} \text{ N/m}$$

35. **Ans (1)**

Due to uniform speed F_{net} on it is zero

which is

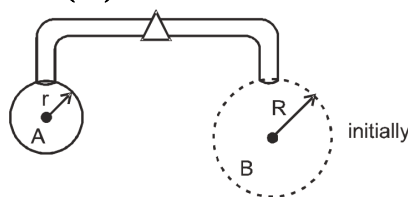
$$v_T \propto r^2$$

$$\frac{v_{T_2}}{v_{T_1}} = \frac{(2r)^2}{r^2}$$

$$v_{T_2} = 4v$$

Now if radius is doubled speed will be four times.

36. **Ans (3)**



$$P_A = P_0 + \frac{4\sigma}{r};$$

$$P_B = P_0 + \frac{4\sigma}{R} \quad \{P_0 = \text{atmospheric pressure}\}.$$

Clearly $P_A > P_B$; so air will flow from A to B.

As r decreases; pressure will become more and hence more flow of air from A to B.

Ultimately bubble A collapses and B becomes bigger in size.

38. **Ans (1)**

$$x = 3t^2 \quad \therefore v = \frac{dx}{dt} = 6t \quad \& \quad a = \frac{dv}{dt} = 6$$

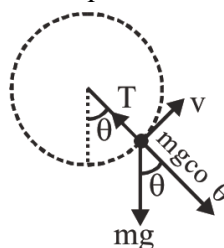
$$P = F \cdot v = m a v = 2 \times 6 \times 6t$$

$$P = 72 \times 4 = 288 \text{ W}$$

Hence option (1)

39. **Ans (2)**

$$T = \frac{mv^2}{r} + mg \cos\theta$$



as θ increases $\cos\theta$ and v

both decreases

hence for $\theta = 60^\circ$, T will be less i.e., $T_1 > T_2$

40. Ans (2)

It is a case of elastic collision. As masses of two balls are equal and 2nd ball is at rest before collision, hence after the collision the speeds are just exchanged, i.e., 1st ball will now come to rest while the second ball will move forward with the same speed as that of the first, i.e., 5 m s^{-1}

41. Ans (2)

$$I = I_{\text{larger}} + I_{\text{smaller}}$$

$$I = 4M(4R)^2 + [MR^2 + M(5R)^2] = 90 MR^2$$

42. Ans (3)

$$\theta = \omega_0 t + \alpha \frac{1}{2} t^2$$

$$200 = 0 + \frac{1}{2} \alpha (5)^2$$

$$\alpha = 16 \text{ rad/s}^2$$

43. Ans (1)

$$\vec{P} = \vec{\tau} \cdot \vec{\omega}$$

$$= (\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (2\hat{i} + 3\hat{j} + 4\hat{k})$$

$$= (2 + 6 + 12) = 20 \text{ watt}$$

44. Ans (4)

$$\text{K.E.} = \frac{1}{2} I \omega^2$$

$$I_{\text{about diameter}} = \frac{MR^2}{2}$$

$$\therefore \text{K.E.} = \frac{1}{2} \left(\frac{MR^2}{2} \right) \omega^2$$

$$= \frac{1}{2} \times \frac{10 \times 0.5 \times 0.5}{2} \times 400$$

$$= 250 \text{ J}$$

49. Ans (4)

$${}_6\text{C} : 1s^2 2s^2 2p^2, \text{ total } e^- = 6$$

$$\text{total IP} = 6$$

52. Ans (2)

$$Z_{\text{eff}}, \text{EN} \Rightarrow \text{N} < \text{O} < \text{F}$$

$$\text{IP} \Rightarrow \text{O} < \text{N} < \text{F}$$

$$\text{P}^3 \text{ conf.}$$

59. Ans (4)

$$\text{I.P.} \propto \frac{+\text{ve charge}}{-\text{ve charge}}$$

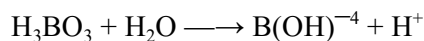
63. Ans (1)

Concept

74. Ans (4)

NCERT-XI, Pg No. 126, Para-2

77. Ans (3)



80. Ans (3)

$$\text{F.C.} = \frac{\text{ch arg e}}{\text{No. of 'O'}} \quad \frac{-1}{1} \quad \frac{-1}{2} \quad \frac{-1}{3} \quad \frac{-1}{4}$$

$$\text{B.O.} = 2(\text{F.C.}) = 1, \quad 2 - (-1/2), \quad 2 - (-1/3), \quad 2 - (-1/4)$$

$$= 1.5 \quad 1.67 \quad 1.75$$

83. Ans (2)

$$\left[\begin{array}{l} \text{Basic nature} \\ \text{of oxide} \end{array} \propto \begin{array}{l} \text{Metallic} \\ \text{character} \end{array} \propto \frac{1}{\text{I.P.}} \right]$$

$\therefore \text{K}_2\text{O}$ – Most basic

84. Ans (2)

tendency of forming complex $\propto$ charge density

$$\propto \frac{1}{\text{size of atom}}$$

86. Ans (3)

Only amphoteric metal react with NaOH

93. Ans (1)

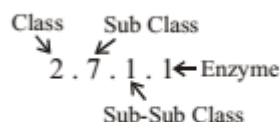
NCERT-XI Pg. # 204

104. Ans (2)

NCERT Pg. # 229

107. Ans (4)

Four digit code of enzymes $\Rightarrow$



Hexokinase is the enzyme of 2nd class transferases so its code number starts with digit '2'

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| <p>126. Ans (4)
NCERT (XIth) Pg. # 211</p> <p>128. Ans (3)
NCERT (XI) Pg. # 230</p> <p>138. Ans (4)
NCERT Pg.# 250</p> <p>139. Ans (1)
NCERT-XI, English Page # 157, (Last Para) and Fig.9.7
हिन्दी पेज # 157, (नीचे से दूसरा पैराग्राफ) एवं पेज # 158 (चित्र 9.7)</p> <p>141. Ans (2)
NCERT, Page No. # 139</p> <p>142. Ans (1)
NCERT (XI) Pg. # 132, Para-1</p> <p>145. Ans (3)
- Root tip cell is '2n'
- $2n = 20$ (acc. to Question)
- Meiocyte is also 2n
Meiocyte ($2n$) $\Rightarrow 20$
- Tetrad is a pair of homologous chromosomes.
So, tetrad = 10</p> <p>146. Ans (3)
NCERT Pg.#125, 126</p> <p>148. Ans (2)
NCERT (XIth) Pg. # 133, fig.-8.6</p> <p>150. Ans (4)
NCERT, Page # 169, Para 10.4.1</p> | <p>153. Ans (2)
NCERT Pg. # 135</p> <p>154. Ans (3)
NCERT XIth Pg.# 163</p> <p>155. Ans (3)
NCERT Pg. # 165,166</p> <p>157. Ans (2)
NCERT XI Pg. # 134</p> <p>158. Ans (1)
NCERT (XIth) Pg. # 138</p> <p>159. Ans (2)
NCERT Page # 166</p> <p>167. Ans (3)
NCERT-XI, Pg. # 146</p> <p>168. Ans (3)
NCERT (XI) Pg # 151</p> <p>170. Ans (1)
NCERT (XIth) Pg. # 148</p> <p>172. Ans (2)
NCERT (XI) Pg # 147</p> <p>177. Ans (3)
NCERT-XI, Pg. # 150, Para-1, Line-1,2,3,4</p> <p>179. Ans (3)
NCERT-XI, Pg. # 146</p> <p>180. Ans (4)
NCERT XII, Pg # 183</p> |
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